

Public Policy Frameworks for Improving Population Health

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ABSTRACT: Four conceptual frameworks provide bases for constructing comprehensive public policy strategies for improving population health within wealthy (OECD) nations. (1) *Determinants* of population health. There are five broad categories: genes and biology, medical care, health behaviors, the ecology of all living things, and social/societal characteristics. (2) *Complex systems*: Linear effects models and multiple independent effects models fail to yield results that explain satisfactorily the dynamics of population health production. A different method (complex systems modeling) is needed to select the most effective interventions to improve population health. (3) *An intervention framework* for population health improvement. A two-by-five grid seems useful. Most intervention strategies are either ameliorative or fundamentally corrective. The other dimension of the grid captures five general categories of interventions: child development, community development, adult self-actualization, socioeconomic well-being, and modulated hierarchical structuring. (4) *Public policy development process*: the process has two phases. The initial phase, in which public consensus builds and an authorizing environment evolves, progresses from values and culture to identification of the problem, knowledge development from research and experience, the unfolding of public awareness, and the setting of a national agenda. The later phase, taking policy action, begins with political engagement and progresses to interest group activation, public policy deliberation and adoption, and ultimately regulation and revision. These frameworks will be applied to help understand the 39 recommendations of the *Independent Inquiry into Inequalities in Health*, the Sir Donald Acheson Report from the United Kingdom, which is the most ambitious attempt to date to develop a comprehensive plan to improve population health.

INTRODUCTION

Copious data, confirmed in practically every study and society examined, has identified with sufficient confidence many of the key social and societal factors that if improved would elevate population health. Further research undoubtedly will broaden, add important insights, and refine the texture of our understanding. Nonetheless, the knowledge base that exists in 1999 is sufficiently comprehensive and

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robust to support the beginning of selected aspects of a population health improvement program.

The improvement of certain societal features would at a minimum improve the general quality of living overall, but would likely improve population health as well. These features include improved opportunities for the following: successful child development, strengthened community cohesion, enhanced self-fulfillment, increased socioeconomic well-being, and modulated hierarchical structuring.

Multipronged actions initiated by multiple sectors are likely to be most effective. The sectors include nonprofit community and national organizations; faith organizations; philanthropies; schools; the recreational, entertainment, and media groups; business; political parties; public policy interests; and local, regional, and national governments. This paper is limited to public policies to improve population health, but the public policies are unlikely to be effective, or even adopted, unless there is in parallel an activation of multiple sectors and synergism is achieved. Social currents, directions, and norms become embedded in expectations, behaviors, and operations. Accepted paradigms ultimately become encoded in laws and regulations. Even relatively modest shifts in social norms, say five degrees out of a whole circumference, will be difficult to achieve. Movement toward more healthful societal circumstances will require multiple approaches and the mobilization of understanding, concern, and commitment of multiple sectors. Public policy development usually does not lead, but rather follows broad public concern.

Four conceptual frameworks, when integrated, can provide guidance for constructing public policy ideas and developing strategies for improving population health within developed nations: (1) determinants of population health, (2) complex system modeling, (3) intervention framework, and (4) public policy development process. The four conceptual frameworks will be described, and then applied to an assessment of the 39 recommendations made in *Independent Inquiry into Inequalities in Health*,¹ the 1998 Sir Donald Acheson report from the United Kingdom, the most ambitious research-based attempt to date to formulate a comprehensive plan to improve population health. Although many chapters in this book advance policy recommendations, the comprehensiveness and coherency of the *Independent Inquiry* provide an opportunity to illustrate the conceptual frameworks for policy developed for this chapter.

DETERMINANTS OF POPULATION HEALTH

There are five major categories of influence on health: genes and associated biology; health behaviors such as dietary habits, tobacco, alcohol and drug use, and physical fitness; medical care and public health services; the ecology of all living things; and social and societal characteristics (FIG. 1). To summarize, the relative proportional influence of each of the five categories is unknown in precise quantitative terms. FIGURE 1 should be interpreted as a crude approximation at this stage of the science. The dashed radii are intended to convey rough estimates, as well as the interdependence/interactivity of the various influences. The absence of a radial line separating total ecology from social/societal characteristics reflects the lack of quantitative knowledge on these two categories of determinants at this time. Genes,

DETERMINANTS OF POPULATION HEALTH

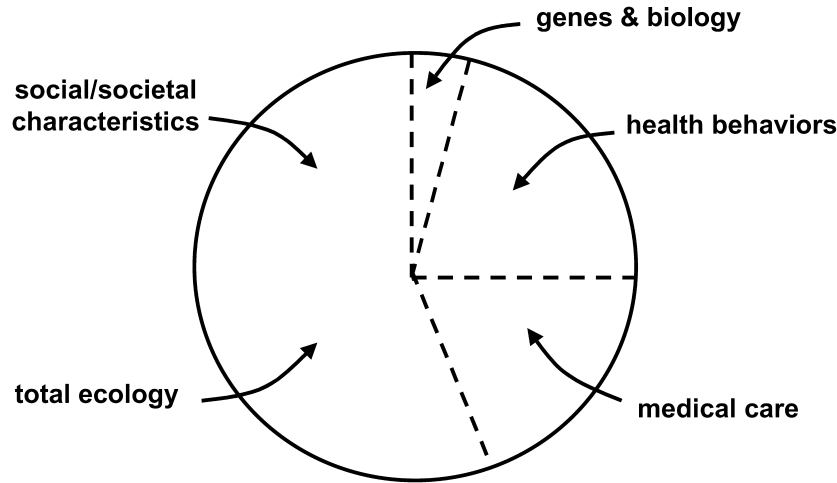


FIGURE 1. Relative influence of the five major determinant categories of population health: rough approximations.

health behaviors, medical care, total ecology, and social/societal characteristics comprise a big, complex, and dynamic network of interactive variables that is understood in a general sense but not understood in a precise, quantitative way. A large body of evidence indicates that social/societal factors exert a major influence on population health. Efforts to improve population health through policies to modify the other four categories of influence while leaving social/societal characteristics unchanged are unlikely to be successful.

COMPLEX SYSTEM MODELING

Mention is made above of the interactivity of most of the factors that influence population health. Changes in one induce responses in the others. Positive and negative feedbacks and cancellation and synergistic effects are predominant features of hugely complex systems exemplified by the influence of surrounding factors on population health. When added to the surrounding factors the physiological systems within the human being that mediate the social effects with uncountable numbers and kinds of adaptations and adjustments, the health production system reveals itself as almost incomprehensibly complex. Linear effects models, multiple independent effects models, and multivariate analytic methods that have driven the social determinants of health field up to this time fail to yield results that satisfactorily explain the dynamics of population health production. Population health production is unlikely to be understood from analyses of individual components. Sociobiologic

system complexity cannot be explained mechanistically or predictably as can the internal combustion engine or chemical equations. The social determinants of health field, and most particularly the ability to predict with greater certainty the multitudinous consequences of interventions, require that the concepts and measures of complex systems be applied. It is noteworthy that 30 pages of a recent issue of *Science*² have been devoted to exploring complex systems related not only to chemistry and the nervous system but also to social systems such as the grouping behavior of animals and the economy.

Yet, despite quantitative shortcomings in our ability to assign precise numerical causal roles to each class of population health determinant and our inability to isolate with precision the impact on population health of each variable in the complex social-health system, there are several broad categories of interventions that could beneficially be applied now. Improvements in child development, community cohesion, self-fulfillment, total ecology, and socioeconomic mobility would result generally in improved quality of life, and at the same time these improvements would likely be salutary to population health. Reasonable evidence, not certainty of knowledge, permitted tobacco control to move forward 40 years ago. Existing data is adequate for formulating policies and other actions that could affect population health importantly. Awaiting new analytic methods and quantitatively more precise information will delay by decades or longer attempts to improve population health.

AN INTERVENTION FRAMEWORK FOR POPULATION HEALTH IMPROVEMENT

The intervention framework (TABLE 1) identifies five broad intervention objectives that are likely to be salutary for population health. The five are improved child development, strengthened community cohesion, enhanced self-fulfillment, increased socioeconomic well-being, and modulated hierarchical structuring; that is, interventions aimed at children, the community, adults, the economy, and arrangements for social positioning. Each intervention can be classified as either ameliorative or fundamentally corrective. For example, ameliorative interventions to improve child development might include approval of a city ordinance that allows surplus space in public school buildings to be used for day care while parents are working, or to reinvigorate the YMCAs and the YWCAs so that supervised after-school recreational activities for children and youths become generally available. Fundamentally

TABLE 1. Intervention framework to improve population health

INTERVENTION OBJECTIVES	AMELIORATIVE	FUNDAMENTALLY CORRECTIVE
Improve <i>child</i> development		
Strengthen <i>community cohesion</i>		
Enhance opportunities for <i>self-fulfillment</i>		
Increase <i>socioeconomic</i> well-being		
Modulate <i>hierarchical</i> structuring		

corrective programs to improve child development might include programs to train fathers and mothers in parenting skills and in establishing home environments conducive to positive cognitive, emotional, and behavioral development, and developing day care programs having high standards, well-trained and culturally diverse professionals who earn professional wages, transportation that makes the program within practical reach of families, and financial foundations to make the programs affordable to all. The intervention framework could help a community or organization develop short- and long-range planning and identify a combination of ameliorative and fundamentally corrective strategies to provide some near-term accomplishments as well as long-term restructuring that addresses the population health program at its roots.

Other examples can be chosen for strengthening community cohesion, enhancing opportunities for self-fulfillment, increasing socioeconomic mobility, or for modulating the effects of hierarchical structuring. The examples are likely to include combinations of public policies, private sector actions, and community programs, and the active involvement of multiple sectors as presented earlier in this chapter. We will return to this intervention framework in reference to the *Independent Inquiry into Inequalities in Health*.

PUBLIC POLICY DEVELOPMENT PROCESS

This framework separates policy development into two phases (FIGURE 2), an initial phase leading to the development of a public consensus and a later political phase when specific policy actions are taken. Before political action, a broad public understanding needs to be acquired that population health problems have origins in real issues that can be addressed remedially to everyone's advantage. Once that understanding has been assimilated, an evident desire must develop at a high enough priority among a sufficient proportion of the population to create a national agenda, or an authorizing environment and momentum for action. When sufficient momentum has developed, the political process will be authorized to pursue policies to address the problem. This framework helps decide where to apply energy in implementing strategies for population health improvement.

In the example of improved child development used above, all aspects of the initial phase have already been accomplished. That is, a public consensus has formed, a national agenda has been developed, and an authorizing environment has developed that will make it natural and acceptable to engage the political process in thinking through alternative proposals to improve opportunities for successful child development. However, although early childhood experiences are commonly known to be related to cognitive, emotional, and behavioral development, it is not well known that the quality of early childhood development is closely tied to adult health. Americans also place a high value on adult health. Therefore, while the issue of child development is ready for political engagement, the policy action phase might be advanced with greater force if the adult health issue is joined.

On the other hand, a plan to elevate socioeconomic well-being that includes a component of income redistribution should acknowledge that the startling rise in income inequality in the United States has been well documented in books and reports

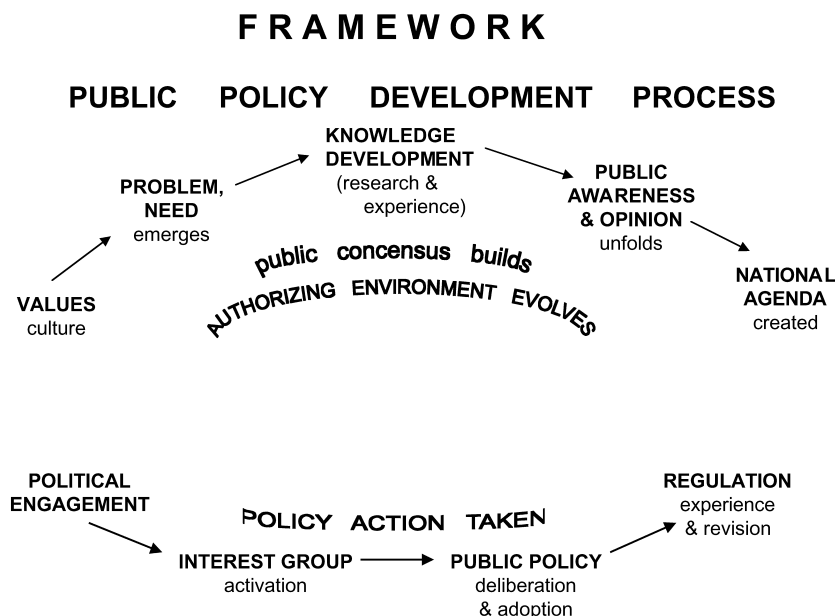


FIGURE 2. Conceptual framework for the two phases of the public policy development process: public consensus/national agenda building, and political/public policy actions taken.

from universities, research organizations, some nonprofit organizations, and journalists. Yet, the relationship of income inequality to gross inequalities in health has not yet stimulated a broad and sustained dialogue in the U.S. media and on the political campaign trails. Nor has the problem risen to occupy a position on the national agenda as has been achieved for issues in education, social security, Medicare, patient bill of health care rights, and international finance and trade. In contrast to the child development issue that is ripe for political action, the income distribution issue should start in the public awareness and public consensus arena. Political productivity is the end game of social transformation.

All four conceptual frameworks should be integrated into a coherent strategy for improving population health. Although the emphasis in this chapter is on public policies, surely corporate policies, community programs, medical care, and health behaviors have important roles to play in child development and adult health. We would lose important potential allies if we ignored the corporate, community, health care, and behavior change public health sectors. Likewise, an understanding of complex systems, even at a low level of sophistication, will be important in anticipating the likelihood of multiple effects of interventions and in maintaining an early alertness and watchful eye for unintended and undesired consequences of interventions. Further, desired outcomes in complex systems can be accomplished through several routes, allowing the selection of an alternate public policy initiative that has a greater

public and political chance of being adopted and sustained. The intervention framework provides the key choices in relatively neutral terms for specific objective setting and intervention concentration, while the public policy development framework can help design the strategy and select the venue in which to initiate the strategy.

INDEPENDENT INQUIRY INTO INEQUALITIES IN HEALTH

The New Labor Party government of Prime Minister Tony Blair, United Kingdom, in July, 1997, requested Sir Donald Acheson, former Chief Medical Officer of the National Health Service, to review and summarize inequalities in health in England, and to identify priority areas for future policy development likely to offer opportunities for government to develop beneficial cost-effective and affordable interventions to reduce health inequalities.

The charge of the inquiry instructed that the policy proposals were to be based on “scientific and expert evidence” and “within the broad framework of Government’s financial strategy” (Ref. 1, pp.155–157). The report was to be completed in about a year, and indeed was submitted to the government in September, 1998.

Several limitations were imposed at the outset. The report was to be focused on government opportunities, as opposed to private sector initiatives. The recommendations were to be framed in the context of the new governments’ financial plans for the country. Only one year was allowed for a project that many of us would have regarded as a three or more year undertaking.

A salient feature of the inquiry is its consistent insistence that the summarization of knowledge be based on science and that the recommendations be supportable by the scientific evidence and by peer review by scientists expert in this field. The process and the report were overseen by a five-member Scientific Advisory Group.

The inquiry was guided by a socioeconomic model of health initially proposed by Dahlgren and Whitehead in 1991 (Ref. 1, pp.5–6). The model emphasizes the context in which we live and by which health or disease is generated. The context is depicted by concentric rings with people at its center enveloped successively by the influences of lifestyle, social and community characteristics, and finally an array of macro-socioeconomic, cultural, and environmental conditions. The influence of this contextual conceptualization, sometimes referred to as socioecologic, is evident in the report.

The report is in two parts. Part 1 contains a summary of research data on inequalities in health. Part 2 consists of reviews of the evidence, amplified from the data cited in Part 1, upon which the policy formulations are discussed. A list of 39 policy recommendations is given at the end of Part 2.

Readers might tend to turn directly to the list of recommendations, but it is a mistake if one’s analysis of the report ends there. Out of context, the recommendations can be interpreted as a war on poverty or as a welfare program for disadvantaged mothers and children. A reading of the entire Part 2, however, adds background, content, depth, and texture to the recommendations. They become a comprehensive, integrated, and plausible set of recommendations for government policies designed to reduce inequalities in population health. The recommendations address population health inequalities induced by health behaviors, by deficiencies in medical care

planning and delivery, or by the pervasive influence of social and societal characteristics. Although the principal emphasis is on social and societal factors, the argument is well made that inequalities in medical services often sustain or amplify inequalities in health.

The Recommendations

Most of the 39 recommendations have multiple sub-recommendations, and many recommendations are cross-listed under several of the 13 recommendation categories devised for the report (TABLE 2). For simplicity in this chapter, the recommendation categories can usefully be collected into four "groupings" (TABLE 3). The groupings will only be scanned briefly here.

Group A, Scope and Emphasis

Recommendations 1 and 2 set a comprehensive tone by indicating that all government policies should be examined for their possible impact on health inequalities, and indicates that the report gives special emphasis to the less well-off with highest (but not exclusive) priority to women of child-bearing age, expectant mothers, and children.

Group B, Sociostructural Improvements

Recommendations 3 through 20 will perhaps be of greatest interest to this book's readers; they are summarized in TABLE 4. Assessment of *all* public policies for their

TABLE 2. Thirteen categories used for the 39 recommendations in the *Independent Inquiry into Inequalities in Health, United Kingdom, 1998*

1. General	8. Mothers, children, families
2. Poverty, income, benefits	9. Young people and workers
3. Education	10. Older people
4. Employment	11. Ethnic minorities
5. Housing	12. Gender
6. Mobility, transport, pollution	13. National health services
7. Nutrition and agriculture policy	

TABLE 3. Four groupings (by the author of this chapter) for the 39 recommendations in the *Independent Inquiry into Inequalities in Health, United Kingdom, 1998*

Group	Number of Recommendations
A. Scope and emphasis	(2)
B. Sociostructural improvements	(18)
C. Disadvantaged emphasis	(16)
D. Health services	(3)

impact on health inequalities appears in TABLE 4 with specific attention to employment and nutrition (agricultural) policies. Income transfers are invoked to lift the bottom out of poverty, to ameliorate the effects of unemployment, and to assure the affordability of wholesome foods for all. Benefits strategies are advanced by the report with respect to expanding preschool opportunities, improving job training, increasing the availability and quality of public housing, and increasing public transport. Again, a full understanding of the sweep of the report should be achieved by reading the texts of both Parts 1 and 2.

Group C, Disadvantaged Emphasis

Recommendations 21 through 36, specify the report's emphasis on mothers, children, and families; young people and workers; older people; ethnic minorities; and young men and young women separately. To cite just a few examples, for families the report recommends elimination of poverty by income transfers, the elimination of food poverty through distribution of surpluses and assuring affordability, greater opportunities for day care and preschool education, and social and emotional support services for parents through increasing the role of "health visitors." The issue of material inequality is addressed for older people through income transfers and benefits, and for ethnic minorities the report recommends that socioeconomic inequalities be reduced. The span of the recommendations for the disadvantaged can be appreciated by reading the full report. A large fraction of the specific recommendations under Groups A, B, and D would also be beneficial to the disadvantaged.

TABLE 4. Recommendations for sociostructural improvements, Group B, 18 recommendations, in the *Independent Inquiry into Inequalities in Health*, United Kingdom, 1998

Needing Improvement	Recommendations
Poverty and income inequality	Income transfers Benefits
Education	Increase funds for preschools and less well-off schools, and expand health promotive schools
Employment	Improve training and job quality Study impact of all employment policies Ameliorate affects unemployment
Housing	Increase availability and quality of public housing
Mobility, transportation, pollution	Increase public transportation Decrease motor vehicle use Lower speed limits Increase cycling and walking
Nutrition	Study impact of agricultural policies Improve distribution surplus Wholesome foods in grocery stores Ensure affordability of foods

Group D, Health Services

Recommendations 37 through 39 seek to promote equity of access and quality of services. The report recommends that resource allocation for health services be differentially determined by needs weighting for each specific population. Monitoring of improved equity should be facilitated by adequate data systems and triennial audits.

A brief summary of the report does not do justice to its expanse. Its objective is to reduce inequalities in health through a reassessment of all government policies that might have a direct or indirect effect on health inequality. It uses all avenues including medical care, preventative public health measures, encouragement of more salutary health behaviors, and a large measure of sociostructural remodeling. The latter includes direct actions for diminishing income inequality (income transfers) and recommends a wide range of expanded benefits intended to reduce inequalities in health. The comprehensiveness of the report's attention to a wide panoply of structurally embedded societal features commands attention by everyone concerned about the recalcitrant problem of health inequalities within societies.

A U.S. PERSPECTIVE (INTERVENTION FRAMEWORK) APPLIED TO THE INDEPENDENT INQUIRY'S RECOMMENDATIONS

The cultural, social, and political contexts of the United Kingdom and the United States are sufficiently dissimilar to justify skepticism that conceptual frameworks for action are cross-applicable. Nonetheless, as scientists and others working in the field of society and population health turn attention to the practical work of fostering development of actual programs and social policies to improve population health, concepts and theories will be needed to guide the formulations and to ground imaginations in reality. Two conceptual frameworks (interventions, public policy development) offered in this chapter are works in progress. There is no empiric evidence of their validity or their practical usefulness. These works in progress might be sharpened and made more useful by applying them to the independent inquiry's recommendations in a test more or less of the validity of the concepts within the frameworks. TABLE 5 is an attempt to do that.

In this depiction we have placed each of the report's recommendations on sociostructural remodeling into the grid of intervention objectives and assigned them as most likely to be in the ameliorative or fundamentally corrective category. Using child development as an example, expanding preschool opportunities for children aged 0–5 and using financial support allocation formulas that are weighted according to the needs of the particular students of each school are both fundamentally corrective. For increasing socioeconomic well-being, income transfers and benefits programs are fundamentally corrective. For enhancing opportunities for self-fulfillment using employment policies, assessing and responding supportively to the effects of unemployment can be ameliorative of a problem that already is in existence, whereas elevating skill levels of workers by institutionalizing training and undertaking a comprehensive assessment of all employment policies regarding their direct and indirect effects on health inequalities could be fundamentally corrective actions.

TABLE 5. Recommendations for sociostructural improvements made by the *Independent Inquiry into Inequalities in Health* (U.K.) placed into the conceptual framework for interventions advanced in this paper (U.S.)

Intervention	Ameliorative	Fundamentally Corrective
Improve <i>child</i> development		EDU: preschools weighted funding
Strengthen <i>community cohesion</i>		MOB: ↑ public transport ↓ motor vehicles ↑ cycling, walking
Enhance opportunities for <i>self-fulfillment</i>	EMP: unemployment effects HOU: public housing NUT: distribute surplus grocery stores	EMP: training/skills policies review
Increase <i>socioeconomic</i> well-being		PII: income transfers benefits
Modulate <i>hierarchical</i> structuring		

ABBREVIATIONS: EDU, education; EMP, employment; HOU, housing; MOB, mobility; NUT, nutrition; PII, poverty, income inequality.

What does the intervention framework reveal about the recommendations of the report? Our interpretations should be regarded as tentative, and perhaps even foolhardy, because of our ignorance of the British value structure, politics, present and long-range currents in social transformation, and the present state of laws and regulations. With reservations, and in the spirit of a desire to understand whether the framework has any utility, two interpretations are offered. First, the report advances relatively few recommendations that are ameliorative, at least with respect to sociostructural modifications. Ameliorative actions respond to the present population's needs and sufferings and in many ways are reflections of a society's empathy and humanitarianism toward its fellow citizens. The empty spaces in the ameliorative column can possibly be explained by the fact that the charge to the inquiry specifically circumscribed the attention to "...government...interventions to reduce health inequalities." Private sector organizations and communities are more likely to take ameliorative actions. Perhaps the report's relatively greater emphasis on fundamentally corrective policies should be lauded, especially in light of the report's recommendation No. 1 that *all* policies be reviewed for their possible impact on health inequalities.

Second, the report offers no recommendations to modulate hierarchical structuring. This might be the most difficult target area to restructure. Most of the research and published attention on social inequality has concentrated on the most easily

measured social variable, that is, per capita or household income. But other elements of hierarchical social structures might be fundamentally and more profoundly causative of health inequalities. Some of these include hierarchically graded distributions within a social structure of status, opportunity, privilege, power, and authority. These variables have not been addressed in the research and have been absent from the public discourse, little as it has been, on social characteristics and population health inequalities.

How do the independent inquiry's recommendations on sociostructural improvements fit into the framework for the public policy development process (Fig. 3)? To reiterate, the conceptualization of the process for the United States is not likely to be transferable to the United Kingdom. However, perhaps something can be learned from doing so.

To begin, all of the recommendations of the report are framed as recommendations for government action because the inquiry was conceived of and framed by the government elected to office at that time. As a result all recommendations enter the process at a late phase of the public policy development process—at the public policy deliberation stage.

I would think that, in the hypothetical exercise of applying the inquiry's report to America, a preferred strategy would be to enter the process at an earlier phase, as

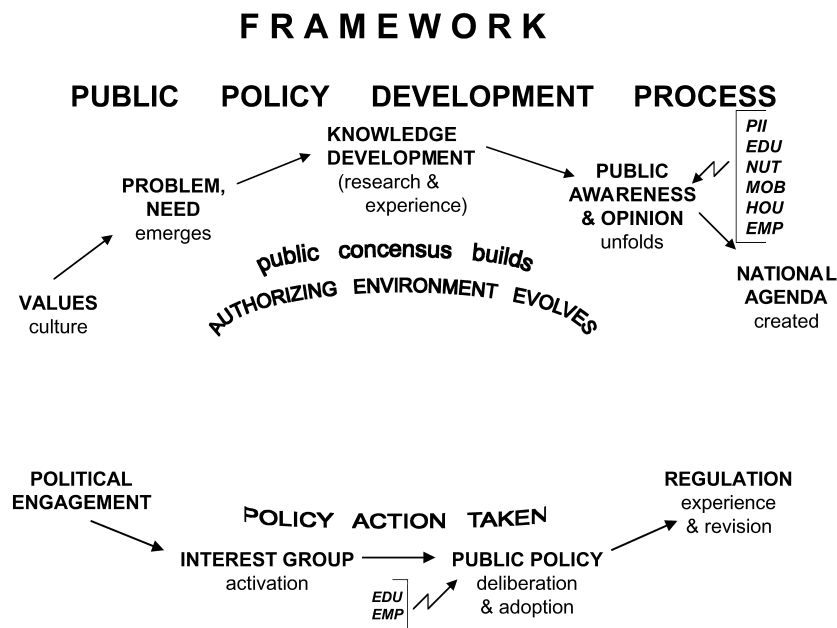


FIGURE 3. Recommendations for sociostructural improvements made by the *Independent Inquiry into Inequalities in Health* (UK) inserted into in the public policy development process in places (*italics*) that the author of this chapter believes would be appropriate for the circumstances of the United States.

depicted in FIGURE 3. Political, media, and public awareness of the causative connection of social position to inequalities in health does not exist in the U.S. The national popular agenda does not include health related to hierarchical structuring except as an issue of poverty. A public consensus on this subject does not exist. An authorizing environment has not been created. Social-health inequalities cannot be engaged as a political issue.

In the United States, political engagement has begun for small parts of the inquiry's recommendations. For example, expanding preschool availability receives consistent attention in policy discussions related both to working mothers and to children being raised in poverty. Several large and influential nonprofit advocacy organizations promote and sustain these issues. Consistent media attention is directed. The environment has evolved, a national agenda has developed, political engagement and interest groups have been activated, and public policy deliberations are taking place.

But the larger agenda, acquisition by the U.S. society of an understanding of the dynamics of health production through sociostructural influences, has not even begun. Therefore, a strategy more likely to succeed in the United States according to this framework and depicted in FIGURE 3 would start with continued research connected to a national program of elevating public awareness to the level of concern that lifts the issue to the national agenda. The strategy for the United States would begin with research to understand just what the American public, by subpopulations, already knows about the relationship of social features to population health. It is probably very little, except as related to poverty. Knowledge would also have to be gained about specific values and beliefs already held by the public that might be sparked into reverberation with the social-health gradient subject. This would provide a focus for a national public information program using multiple channels of communication and education.

The British report, *Independent Inquiry into Inequalities in Health*, given its purpose, is a highly valuable and progressive source of information and recommendations. It sets out a comprehensive policy agenda to improve population health, at least for developed nations. With individual nations building their own conceptual frameworks to select social restructuring targets for intervention, and with knowledge of the public attitude development process in their country and with approaches that activate the public sector, the private sector, and community action, the independent inquiry provides a treasure of summarized knowledge and comprehensive approaches that other nations will find valuable.

The conceptual frameworks developed in this chapter with some modifications might be salient for health improvement planning at the national, state, or smaller geopolitical unit level. The independent inquiry from the United Kingdom sets a useful example by its comprehensiveness and its attention to the need for sociostructural revision if meaningful population health improvement is to be attained.

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